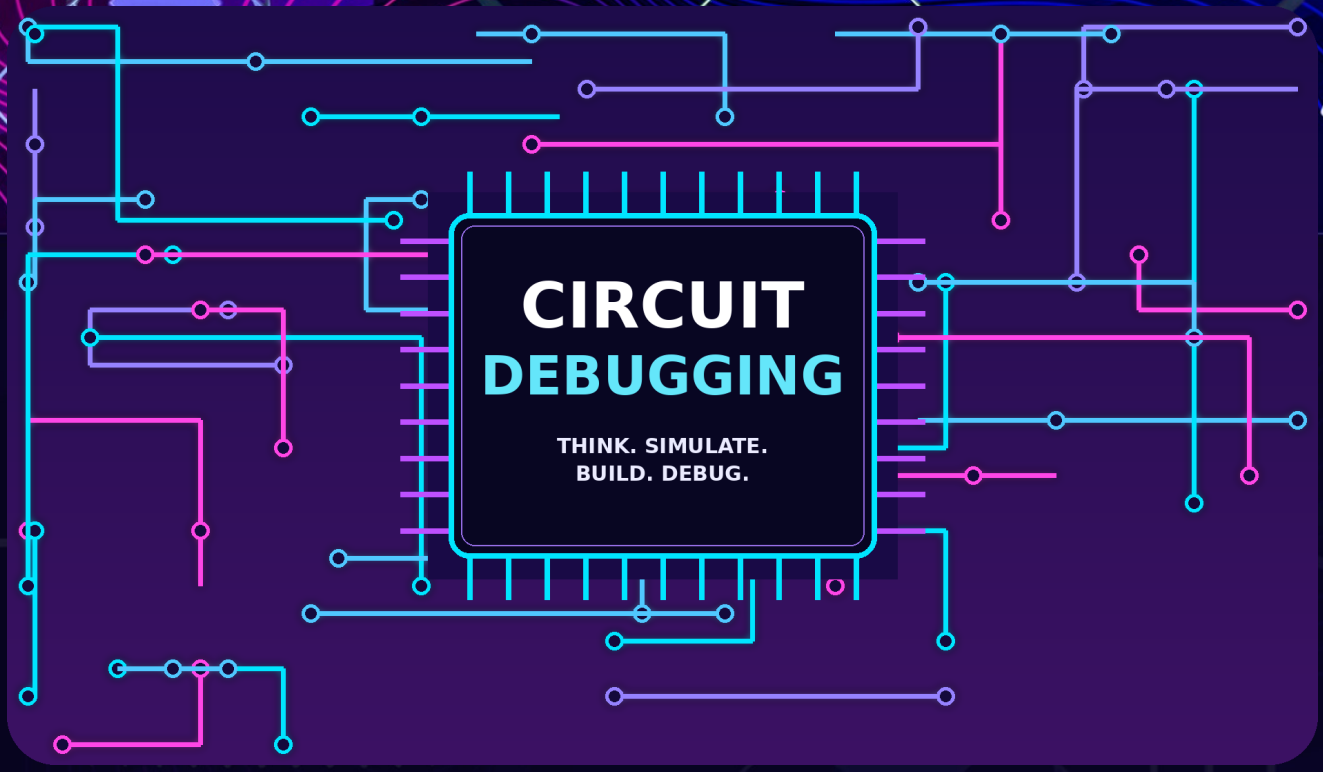
NIT ARUNACHAL PRADESH

ADDOVED

ENTER THE ARENA



Event Overview

Circuit Debugging is a three-round technical event designed to test participants' knowledge, analytical ability, simulation skills, circuit construction, measurement, and practical troubleshooting skills.

The event follows a progressive structure:

Round 1 → Understand

Round 2 → Simulate & Analyse

Round 3 → Debug & Repair

Participants will progress from theoretical circuit analysis to computer-based simulation and finally to real-world hardware debugging.

Team Size

- Each team may consist of a **maximum of 2 participants.**
- Both team members are expected to participate actively throughout the event.
- No team member may be replaced after the commencement of the event unless permitted by the organizers under exceptional circumstances.

NIT ARUNACHAL PRADESH

ADDOVED

ENTER THE ARENA

Event Structure

The event will consist of three rounds:

Round	Name	Mode	Maximum Marks
Round 1	Circuit IQ	MCQ	**30**
Round 2	Simulation Challenge	LTspice	**30**
Round 3	Hardware Debugging	Breadboard	**40**
Total			**100**

The final ranking will be based on the combined score obtained across all three rounds, subject to the qualification criteria specified by the organizers.

ROUND 1 — CIRCUIT IQ

Mode: MCQ-Based Screening Round

Round 1 will test the fundamental knowledge and circuit-analysis ability of the participants.

The questions may cover:

- Basic circuit theory, Ohm's Law and Kirchhoff's Laws
- Series and parallel circuits
- Voltage and current calculations
- Resistors and capacitors
- Diodes and transistors
- Basic electronic circuits, circuit symbols and configurations
- Basic fault identification
- Circuit-based numerical questions

Questions may include both **conceptual and circuit-based MCQs**.

Format

- **Total Questions:** 20
- **Duration:** 20 minutes
- **Total Marks:** 30
- Each question carries **1.5 marks**.
- There will be **no negative marking**, unless otherwise announced by the organizers.

NIT ARUNACHAL PRADESH

ADDOVED

ENTER THE ARENA

Qualification

Based on the number of participating teams, the organizers will determine the number of teams qualifying for Round 2.

The qualification criteria will be announced before the commencement of the event.

ROUND 2 — SIMULATION CHALLENGE

Mode: LTspice Simulation

In Round 2, each qualified team will be provided with a problem statement describing a particular electrical/electronic circuit requirement.

Participants must analyze the given problem, determine the appropriate circuit configuration and design and simulate the required circuit using LTspice within the specified time limit.

Participants will be required to:

1. Read and understand the given problem statement.
2. Construct the circuit in LTspice.
3. Run the appropriate simulation.
4. Obtain the required voltage/current values or waveforms.
5. Submit or demonstrate the required simulation results to the event coordinators.

Note: *Each team must bring their own laptop with LTspice pre-installed and tested before the event.*

Participants are responsible for ensuring that LTspice is installed and functioning properly on their laptop before the commencement of Round 2.

The organizers will **not be responsible for delays caused by software installation, downloading, updating, or laptop-related technical issues during the round**.

ROUND 3 — HARDWARE DEBUGGING CHALLENGE

Mode: Real-Time Breadboard Debugging

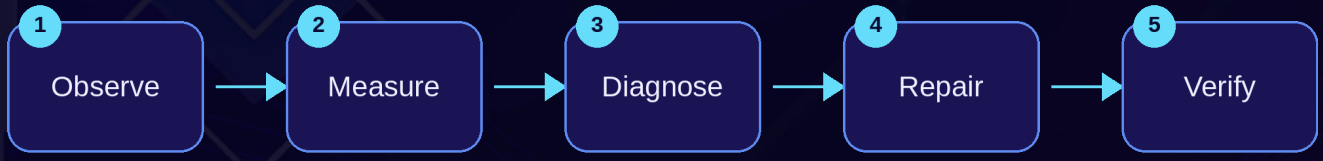
Round 3 will be the final and most practical round of the event.

Qualified teams will be provided with a pre-assembled circuit on a breadboard. The circuit will be designed to perform a particular function but will contain one or more intentionally introduced faults.

The participants will not initially be informed about the location or nature of the fault.

Participant's Task

Teams must systematically troubleshoot the circuit by following:



Participants will be required to:

1. Understand the expected behaviour of the circuit.
2. Analyze the symptoms provided by the organizers.
3. Develop a systematic troubleshooting approach.
4. Use the permitted measuring instruments.
5. Measure voltage, resistance, continuity, or other relevant parameters.

6. Identify the faulty component or connection.
7. Explain the reason for the malfunction.
8. Correct the fault.
9. Demonstrate that the circuit has returned to its intended working condition.

Participants should diagnose the fault before replacing components. **Random replacement of components may not be considered a valid debugging approach.**

General Rules

1. Each team can have a maximum of **2 participants**.
2. Participants must report to the venue before the prescribed reporting time.
3. Participants must carry a valid **college/institute ID card**.
4. **LTspice must be pre-installed and tested on the participant's laptop before the event.**
5. The organizers will provide the required components, breadboards, measuring instruments and other equipment required for the physical rounds.
6. Participants must use the equipment provided by the organizers carefully and responsibly.
7. Participants must not intentionally damage or misuse any equipment, component, breadboard or instrument.
8. Participants must not interfere with another team's circuit, equipment or workstation.
9. Participants must follow all instructions provided by the event coordinators and volunteers.
10. Participants must stop working immediately when the allotted time expires.
11. Any attempt to gain an unfair advantage or obtain unauthorized assistance may result in disqualification.
12. Participants may be asked to explain their calculations, circuit design, simulation or troubleshooting procedure.
13. The organizers reserve the right to inspect LTspice files, calculations, circuit connections and simulation results.
14. Any deliberate misconduct, cheating, equipment damage or interference with another team may lead to immediate disqualification.
15. The organizers reserve the right to modify the sequence, timing or qualification criteria if required due to operational constraints. Any such changes will be communicated to participants before the concerned round.
16. The decision of the **judging panel/event coordinators shall be final** in matters related to evaluation, scoring and qualification.

17. Prize Pool

Prize Value: ₹13,000+ (including goodies)

NIT ARUNACHAL PRADESH

ADDOVEDI

ENTER THE ARENA

Registration Charge

REGISTRATION FEE

₹ 50 per member

Contact Details

EVENT COORDINATOR

Neeraj Bhandari

Ph-7667058211

Join our WhatsApp group for all the future updates!

CIRCUIT DEBUGGING(ADDOVEDI'26)

WhatsApp group

Best Wishes to All Participants!

CIRCUIT DEBUGGING — Observe. Measure. Diagnose. Repair. Verify.

ADDOVEDI'26 . NIT ARUNACHAL PRADESH